CLOUD & DEVOPS

Disaster Recovery Visual Guide

RTO, RPO, and the four strategies — from backup to hot multi-site.

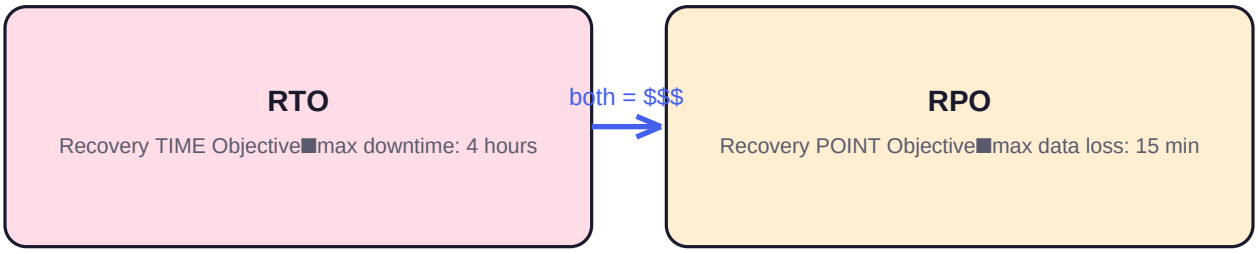
ANIMATED WALKTHROUGH

Step through pages to play the animation

STEP 1/6

RTO & RPO: The Two Numbers

How fast must you recover, and how much data can you lose?



Timeline: disaster at T → data restored to T–RPO → service back by T+RTO.

Set RTO/RPO per tier: checkout gets 1h/5min; the internal wiki gets 48h/24h. Not everything is tier-1.

STEP 2/6

Strategy 1: Backup & Restore

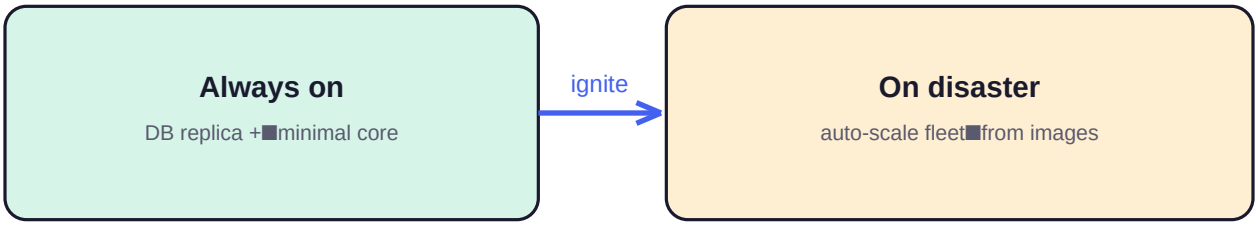
Cheapest. Slowest. Better than nothing.

- RTO: hours. RPO: up to 24h (last backup). Cost: near zero idle.
- Automate the restore — a manual 47-step wiki is not a recovery plan.
- Test restores quarterly: untested backups are Schrödinger's backups.

STEP 3/6

Strategy 2: Pilot Light

Core data replicates; servers spin up on demand.



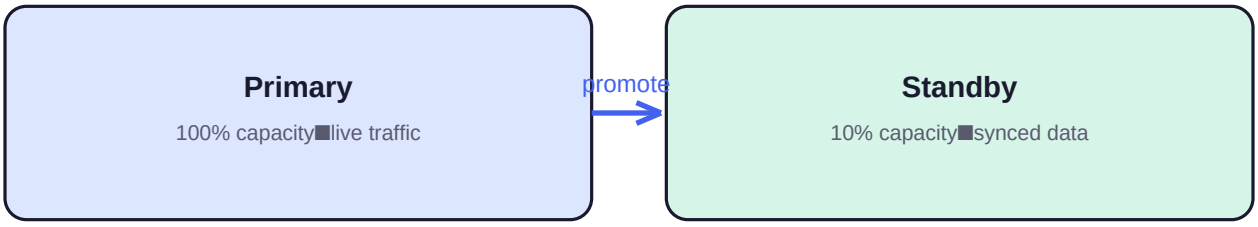
Like a pilot light on a furnace: tiny flame always burning, full heat in minutes.

- RTO: tens of minutes. RPO: minutes (async DB replication).
- Keep golden images / IaC current — stale images fail at the worst moment.
- DNS flip moves traffic once the fleet is healthy.

STEP 4/6

Strategy 3: Warm Standby

Scaled-down clone always running, ready to scale up.



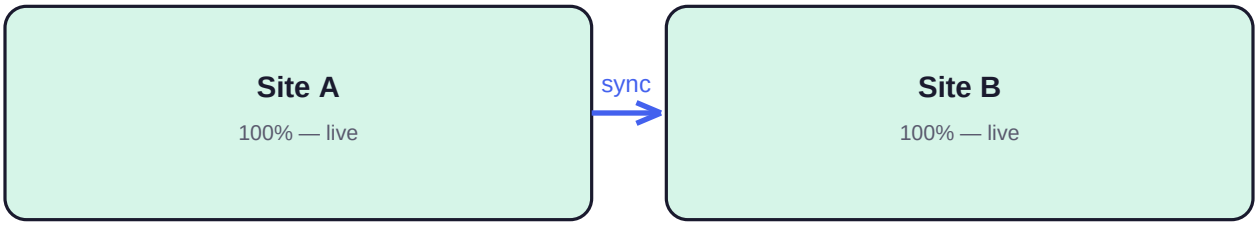
RTO: minutes (scale up + DNS). RPO: seconds–minutes. Cost: ~15–25% of primary.

The standby must take REAL (synthetic) traffic — an idle standby's first real request is its first test.

STEP 5/6

Strategy 4: Multi-Site (Hot-Hot)

Full capacity in two places, both serving.



RTO ≈ 0, RPO ≈ 0. Cost: 2x everything + data-sync engineering.

- This is active-active multi-region — the DR strategy with no failover step.
- Data conflicts are the price: design for multi-writer from day one.
- Only tier-1 workloads justify it — everything else gets warm standby or pilot light.

STEP 6/6

DR Testing: Prove It Works

A plan you've never run is a hope, not a plan.

Measure actual RTO/RPO in every test and compare to targets — then fix the gap before the real thing.

STEP ✓I✓

Key Takeaways

Disaster Recovery — Remember

- RTO = max downtime; RPO = max data loss. Set them per tier, not per company.
- Backup/restore: cheapest, slowest. Pilot light: DB replicates, fleet ignites.
- Warm standby: scaled-down clone, minutes to promote. Multi-site: hot-hot, ~zero RTO.
- Automate restores and failovers — manual wikis fail under pressure.
- Test on a ladder: tabletop → simulated → live. Measure real RTO/RPO every time.

Keep exploring

Pair this guide with the matching video walkthroughs
and hands-on labs at interstitiumlabs.dev